SAME EVIDENCE, DIFFERENT SKILL: STATE-REGENERATION INSTABILITY IN SELF-EVOLVING AGENTS

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ABSTRACT

Self-evolving agents convert execution feedback into persistent skills, but most studies emphasize which trajectories reach the updater rather than whether the updater reliably materializes a useful persistent state. We study a narrower failure mode: byte-identical learner-visible evidence can fail to regenerate the same behaviorally useful skill. A controlled DeepSeek-V4-Pro study of 48 matched evidence pairs, 96 learned states, and 1,728 held-out evaluations first shows that replacing winner evidence with a mixed rejected-witness view has a small heterogeneous mean effect of 0.023, with a frozen confidence interval crossing zero. We then examine one outcome-selected development stream. A historically strong First-Fail state remains directionally useful across frozen-state actor remeasurements, whereas two fresh updater states produced from reconstructed byte-identical First-Fail evidence do not reproduce the historical advantage. These observations rule out evidence disappearance and weaken a pure one-rollout actor-luck explanation, while remaining consistent with a persistent-state generation bottleneck; they do not identify a population variance component. We therefore treat state generation as an experimental variable and define a prospective typed compiler intervention that maps the same trajectory+score package to a finite repair representation and canonical skill state. A fresh balanced Evidence×Generator bridge makes the generator-factor main effect primary across Winner and First-Fail-4 evidence, while separately testing generic/scope canonicalization, same-evidence state-realization disagreement, and any rejected-source moderator. The present paper establishes local state-regeneration instability and a falsifiable generator hypothesis; it does not yet claim that the compiler improves utility or that rejected failures are uniquely beneficial.

1 INTRODUCTION

Self-evolving agents adapt without updating model weights by converting execution experience into persistent procedural state. Recent systems already show that failed trajectories, success/failure contrasts, execution-grounded repair, and accumulated knowledge can improve later skill use (Liu et al., 2026b; Chen et al., 2026; Liu et al., 2026a; Tang et al., 2026). Most evaluations, however, emphasize *which evidence* reaches the updater. They largely treat the state-writing step itself as an implementation detail.

We study a narrower failure mode: the same learner-visible evidence can fail to regenerate the same behaviorally useful persistent skill. This matters because persistent skill text is itself an intervention on future behavior. Two updater calls can receive identical evidence yet synthesize states that differ in scope, redundancy, ordering, specificity, or procedural commitments. Thus evidence selection and state materialization are experimentally separable even when they are implemented by the same LLM pipeline.

Our starting point is a controlled DeepSeek-V4-Pro study of persistent spreadsheet skills. A 48-pair experiment shows that replacing winner evidence with rejected-witness evidence has only a small heterogeneous average effect, so the paper cannot be about universally superior failure feedback. In one outcome-selected development stream, however, a historically strong First-Fail state remains directionally useful when its bytes are frozen and re-evaluated. Reconstructing the learner-visible First-Fail evidence byte-for-byte and invoking the same free-form updater twice does *not* recreate that historical advantage. The evidence remained fixed; the useful state did not reliably reappear.

We organize the pipeline as

$$\text{trajectories} \rightarrow E \rightarrow G \rightarrow S \rightarrow Y, \tag{1}$$

where E selects the learner-visible evidence package, G generates persistent state, S is the realized skill artifact, and Y is future utility. This factorization is bookkeeping rather than a standalone novelty claim. Its purpose is to make the realized persistent state an explicit experimental object. The completed evidence establishes only a *local state-regeneration instability*: one selected case in which identical evidence does not reliably regenerate the historical useful state. It does not estimate a population variance component or show that updater variance dominates actor noise.

This observation motivates a controlled intervention on G. We replace unconstrained state writing with a small typed compiler that maps the same trajectory+score package to a finite diagnosis/repair representation and a canonical skill state. The compiler is not presented as a universal diagnosis algorithm: SkillRevise, SkillCAT, WikiSkill, and related systems already cover substantial parts of trajectory diagnosis, contrastive feedback, and persistent knowledge construction (Liu et al., 2026a; Chen et al., 2026; Tang et al., 2026). Its role here is causal control: remove free-form state-synthesis degrees of freedom while preserving learner-visible evidence.

The prospective bridge therefore separates four questions that earlier versions conflated. The primary question is the *generator factor*: averaged equally across predeclared Winner and First-Fail-4 evidence cells, does constrained generation outperform the native free-form updater? A second, explicitly FF4-specific question asks whether the compiled state survives score-only and diagnosis-cardinality-informed trajectory-text-blind controls; this classifies the substance of the FF4 typed-diagnosis result and cannot explain the Winner-side contribution to the balanced main effect. A third same-evidence repeated-synthesis probe asks whether distinct FF4 FREE state realizations show observed cross-state disagreement beyond re-running the actor on frozen states; a stronger success-propensity interpretation is allowed only under an explicit conditionally iid/independent stationary actor model. First-Fail superiority and the Evidence$\times$Generator interaction are a fourth, parked moderator question. These gates are orthogonal: failure of a secondary interpretation cannot manufacture or erase the primary generator-factor result.

Contributions.

- **Matched-evidence regeneration audit.** In one controlled selected case, a historically useful state remains behaviorally active when frozen, while byte-identical evidence does not reliably regenerate its advantage through the native updater.
- **Bounded causal localization.** Together with a 48-pair heterogeneous evidence study and a failed witness-selector pilot, this rules out evidence disappearance and weakens a one-rollout actor-luck explanation while explicitly leaving selected-case bias and population variance unresolved.
- **Controlled generator intervention and falsification design.** We define a typed canonical compiler, strong trajectory-blind generic controls, SHA-based state-identity handling, and a fresh balanced Evidence$\times$Generator bridge whose primary generator-factor gate is separated from diagnosis, realization, and rejected-source interpretation. Its STOP rules prevent E3, another backbone, or another benchmark from rescuing a failed generator result.

2 Problem Setup and Closest Work

2.1 Persistent skill self-evolution

For an update stream u, the agent executes eight tasks. Task i produces a frozen search pool $P_i = \{\tau_{ij}\}_{j=1}^{K}$ with binary verifier scores $r_{ij} \in \{0, 1\}$. A serving rule chooses an acting winner for the user-facing task, while an evidence projection E determines which trajectory+score package is exposed to the persistent learner. The state generator G maps the eight learner-visible evidence units and initial skill S_0 to a new persistent skill,

$$S = G(E(P_1), \dots, E(P_8); S_0). \tag{2}$$

A frozen downstream actor then executes held-out task x conditioned on S, producing utility $Y(x; S) \in \{0, 1\}$ under the deterministic spreadsheet verifier.

This decomposition creates three distinct estimands. The *evidence-package effect* compares different E while holding the generator fixed. The *state-generation-method effect* compares different G on the same selected evidence. The *state effect* holds the state bytes fixed and remeasures downstream behavior. These are not interchangeable. In particular, a useful frozen state does not imply that the updater can reliably regenerate it, and an evidence source that sometimes yields a useful state does not imply that the evidence source is generally superior.

2.2 Controlled spreadsheet substrate

Our mechanism studies use a locally generated SpreadsheetBench-compatible suite with six predeclared failure families: input/output contract, target sheet/range, schema/key alignment, aggregation/join, formula/materialization, and multi-step pipeline. Each task has deterministic initialized and golden workbooks and a deterministic verifier. This suite is used for causal control, not as a public benchmark. SpreadsheetBench itself contains real-world spreadsheet manipulation tasks and OJ-style evaluation (Ma et al., 2024); external benchmark claims are deliberately deferred until the internal mechanism survives.

The primary actor and updater model in the current mechanism line are DeepSeek-V4-Pro under a frozen Ark route. Actor decoding uses temperature zero, thinking disabled, and no provider retry. The first-party persistent updater is the MindMemOS SkillEvolver path used throughout the controlled study. Exact model routes, skill hashes, trajectory hashes, budgets, and task IDs are bound in execution receipts rather than inferred from manuscript prose.

2.3 Closest work and residual scientific object

Failure-aware feedback and skill evolution. Rethinking Self-Evolving Agent Skills varies the feedback presented to a skill optimizer and shows that failure-containing feedback is often selected, but the ranking of Normal, Fail-only, and Success-only conditions depends on model and benchmark (Liu et al., 2026b). This directly rules out a novelty claim that failures can help. Our residual question is different: with evidence bytes fixed, can the state-writing step itself become an unstable bottleneck?

Contrastive trajectory diagnosis. SkillCAT samples multiple trajectories, contrasts same-task successes and failures, extracts candidate corrections, and validates patches before merging (Chen et al., 2026). It therefore already covers success/failure pairing and contrastive causal extraction. Our First-Fail evidence arm is not presented as a new contrastive-learning idea. We instead use matched evidence arms to factor evidence selection from the generator that materializes persistent state.

Execution-grounded skill revision. SkillRevise diagnoses skill defects from execution traces, retrieves reusable repair principles, and applies execution-anchored revisions that are re-executed and selected by utility (Liu et al., 2026a). Consequently, “trajectory $\rightarrow$ structured diagnosis $\rightarrow$ skill edit” is not our contribution. Our typed compiler is scientifically useful only if it tests a different claim: whether replacing unconstrained state synthesis with canonical compilation changes state reproducibility or downstream utility under the same evidence.

Persistent knowledge accumulation. WikiSkill separates raw execution experience, an accumulated persistent wiki, and executable skills, and uses the knowledge store to guide future skill evolution (Tang et al., 2026). We therefore do not claim novelty for compiling experience into persistent knowledge. The missing object here is narrower: the realized persistent state as a causal intermediate whose variability can be studied independently of evidence selection.

Residual novelty boundary. The paper’s novelty target is therefore not a new feedback source, diagnosis vocabulary, memory store, or validation-gated text editor. It is the controlled matched-evidence study of state regeneration itself: a frozen-state audit asks whether behavior follows persistent state identity rather than one actor draw, and the prospective bridge varies the complete state-generation method while holding learner-visible evidence fixed. The existing case is consistent with a state-generation bottleneck but does not identify it at population scale. If the fresh generator contrast fails, the method-positive story stops. If the compiler helps but First-Fail evidence does not, the failure-learning claim is dropped while the state-generation-method result may remain.

3 From Evidence Heterogeneity to a State-Regeneration Anomaly

We use a sequence of interventions that progressively hold more of the pipeline fixed. The goal is not to turn one selected positive case into a general claim; it is to determine which layer remains plausible after simpler explanations fail.

3.1 Evidence-source effects are small and heterogeneous at scale

Our closed DeepSeek V2 study contains 48 matched pairs, 96 learned states, and 1,728 held-out task evaluations. The two primary states differ in the evidence projection exposed to the same frozen updater. Across pairs, the mean MRW$-$WIN-C utility difference is $+0.0231$, the median is $+0.0139$, and the standard deviation is 0.0780. The frozen one-sided sign-flip test gives $p = 0.1719$, and the bootstrap confidence interval is $[-0.0185, 0.0660]$.

This study therefore does not establish rejected-witness superiority. Individual streams span positive, null, and negative effects, including values from -0.125 to $+0.181$. The appropriate inference is heterogeneity, not a failed attempt that can be repaired by silently adding samples or widening a threshold.

3.2 Selecting a more "diagnostic" failure does not solve the problem

Post-hoc development analysis suggested that progress-related features correlated with pairwise effects, motivating a single-stream witness-selector pilot on `e1-tsr-00`. This stream was chosen after outcome inspection and is therefore mechanism development only. We froze four evidence arms: WIN-C, First-Fail, Progress-Fail, and Progress-Contrast. On 18 held-out tasks they score 13/18, 17/18, 14/18, and 14/18 respectively.

Although the historical First-Fail state is strong, the preregistered purpose of this pilot was to test whether a progress-based selector provides a stable repair rule. It does not. The frozen verdict is `S1_SIGNAL_FAIL_STOP_NO_S2`. We therefore reject the simple story that "choose a later or more diagnostic failure" explains the earlier heterogeneity.

3.3 The strong persistent state remains useful when frozen

The next intervention holds the persistent state bytes fixed and re-runs only downstream evaluation. The historical strong First-Fail state is compared with the corresponding WIN-C state in two fresh complete actor replicates on the same 18-task development panel. Replicate 1 gives 15/18 versus 14/18; replicate 2 gives 16/18 versus 12/18.

These remeasurements do not make the selected stream representative, and they do not prove that First-Fail evidence caused the state. They do weaken a simpler explanation: the original 17/18 First-Fail result was not merely one lucky downstream actor realization attached to an otherwise inert state. The frozen artifact itself carries behaviorally relevant information.

3.4 Byte-identical evidence fails to regenerate the same advantage

We then reconstruct the exact S1 First-Fail evidence rendered to the updater. All 16 rendered packet hashes match the historical inputs. From this byte-identical evidence we create two new free-form updater realizations, then evaluate each fresh state against a fresh WIN-C realization on the same 18 tasks. Replicate 1 is 15/18 versus 15/18; replicate 2 is 11/18 versus 12/18. The mean fresh contrast is -0.0278.

The contrast with frozen-state remeasurement is the key anomaly. Holding the strong state fixed preserves a directional advantage; holding the evidence fixed but regenerating the state does not. Because the fresh updater states and their downstream evaluations both vary, this is not a variance-component decomposition and does not by itself isolate the updater from actor noise. It nevertheless

Table 1: Evidence ladder. The independent unit and scientific interpretation change across stages; no row is promoted beyond its design.

Stage	Held fixed	Observation	Supported interpretation
V2 48-pair study	actor/updater configuration, task design	mean MRW−WIN-C = +0.0231, CI crosses 0	evidence-source effect is heterogeneous
S1 selector pilot	selected stream, updater path	13/18, 17/18, 14/18, 14/18	simple progress selector fails
Frozen-state remeasure	persistent state bytes	FF: 15/18, 16/18; WIN-C: 14/18, 12/18	strong state has reproducible local behavioral value
Exact-evidence replay	learner-visible evidence bytes	FF: 15/18, 11/18; WIN-C: 15/18, 12/18	same evidence does not reliably regenerate strong state

rules out evidence disappearance as a sufficient explanation of the strong historical state and motivates a direct frozen-state regeneration audit.

3.5 What the completed evidence supports

The current evidence supports the following bounded statement:

> In one controlled development case, identical trajectory evidence did not reliably regenerate the historical behaviorally useful persistent skill, while the previously generated strong state retained downstream utility when frozen. The observation is consistent with persistent-state generation as a candidate bottleneck and cannot be explained by evidence disappearance alone.

It does *not* establish that rejected failures are generally better than winners, that free-form updater variance dominates actor variance in the population, or that deterministic compilation is already superior. These stronger claims require prospective interventions rather than retrospective interpretation of the selected case.

4 Typed State Compilation as a Controlled Generator Intervention

The state-regeneration result motivates an intervention on G, the state-generation interface. The compiler is intentionally modest: it is designed first to remove free-form synthesis degrees of freedom under matched evidence, not to claim a universal agent-repair algorithm. It receives the same learner-visible trajectory text and selected binary verifier score as the corresponding free-form updater arm, but replaces open-ended state writing with a finite diagnosis and canonical compilation path.

4.1 Design requirements

A valid generator intervention must satisfy four requirements.

R1: Information parity. The compiler may read only the selected trajectory text and score exposed to the corresponding free-form arm. It cannot condition on the experimental arm, projection name, stream/family label, task family, golden answer, held-out result, historical S1 outcome, or hidden provenance. SHA identifiers are retained only for integrity receipts.

R2: Explicit intermediate representation. The method exposes what it inferred before changing the skill. Otherwise a different free-form prompt simply moves the uncontrolled synthesis step. We represent each evidence unit by

$$d = (\text{failure stage}, \text{failed invariants}, \text{observed evidence}, \text{required repairs}). \tag{3}$$

R3: Canonical state construction. Identical diagnosis bundles must produce byte-identical persistent skills. Duplicate evidence must not increase state length, primitive ordering is fixed, and redundant repair surfaces are removed deterministically.

R4: Generic and scope falsification. A positive trajectory-conditioned result must beat controls that receive no trajectory text but are allowed the same score pattern, and—for the stronger diagnosis claim—the same rendered repair-block count. This prevents a shorter or less interfering state from being mistaken for correct semantic diagnosis.

4.2 Typed diagnosis from learner-visible evidence

The current intervention uses three repair primitives:

$$\mathcal{R} = \{\texttt{VERIFY\_OUTPUT}, \texttt{COMPLETE\_WORKFLOW}, \texttt{RECOVER\_TOOL\_ERROR}\}. \tag{4}$$

For a selected failed trajectory, visible signals map to repairs as follows:

- no materialized write or output save → `COMPLETE_WORKFLOW`;
- output saved but no visible reload/verification → `VERIFY_OUTPUT`;
- visible tool error without visible clean recovery → `RECOVER_TOOL_ERROR`.

A selected successful trajectory is not automatically labelled as needing a generic repair. Multiple repairs may compose.

These rules are deliberately interpretable and narrow. Completion, verification, and recovery are also plausible generic spreadsheet hygiene and closely aligned with the deterministic evaluator. We therefore do not present this vocabulary as standalone algorithmic novelty. Its scientific role is to instantiate a controlled generator whose representation and rendering behavior are fully specified.

4.3 Canonical compiler

For diagnoses $D = \{d_i\}_{i=1}^{8}$ and base skill S_0, compilation computes

$$C(D, S_0) = S_0 \oplus \mathrm{Canon}\left(\bigcup_i d_i.\texttt{required\_repairs}\right). \tag{5}$$

Canonicalization uses a fixed primitive order and deduplicates repeated requirements. `COMPLETE_WORKFLOW` already includes save, reload, and target verification, so a separately rendered `VERIFY_OUTPUT` block is omitted when both are requested. `RECOVER_TOOL_ERROR` composes after completion semantics. Every diagnosis bundle and compiled skill is content-addressed.

The zero-provider implementation is qualified for deterministic compilation, duplicate suppression, hidden-label exclusion, and the three diagnosis routes. For diagnoses corresponding to the manually frozen mechanism interventions, it reproduces the G1 verification, G2 completion, and G3 completion+recovery skill surfaces byte-for-byte. This is implementation qualification, not evidence that those states improve future utility.

4.4 Two trajectory-blind generic controls

Score-only generic maximum. `SCORE_ONLY_GENERIC_MAX` receives the same ordered eight binary scores but no trajectory text. If all scores are one, it returns S_0. If any score is zero, it emits the maximal nonredundant generic v1 bundle, `COMPLETE_WORKFLOW+RECOVER_TOOL_ERROR`. It tests whether simply knowing that failures occurred plus injecting the strongest generic checklist explains the effect.

Diagnosis-cardinality-informed generic maximum. The maximal generic bundle can be longer or broader than a trajectory-conditioned compiler state. We therefore additionally freeze `SCOPE_MATCHED_GENERIC_MAX`. It receives the same eight scores and exactly one scalar from the trajectory-conditioned compiler: the number $k \in \{0, 1, 2\}$ of nonredundant repair blocks that canonical compilation would render. It receives neither trajectory text nor primitive identity. A frozen generic mapping returns S_0 for $k = 0$, a one-block generic completion state for $k = 1$, and completion+recovery for $k = 2$.

The scalar k is itself trajectory-derived through the compiler's diagnosis cardinality, so this is not a pure trajectory-independent or exact-token-length control. It is trajectory-text-blind but diagnosis-cardinality-informed. If FF4_COMP beats the score-only maximum but not this control, the result supports state scope/sparsity canonicalization at most; if it beats both, the result supports benefit beyond rendered repair-block cardinality but still does not isolate semantic diagnosis from every diagnosis-derived variable.

4.5 What the generator comparison identifies

The free-form and compiler paths are complete state-generation methods. They differ in representation, canonicalization, state scope, redundancy, and use of a generative writer. Thus a positive COMP$-$FREE contrast is a *complete generator-method effect*, not an isolated effect of determinism. State length, delta tokens, rendered block count, and skill SHA are recorded as mechanism diagnostics rather than post-hoc matched away.

A separate repeated-synthesis probe is needed for the stronger regeneration/reproducibility interpretation. It repeats the same free-form evidence package exactly once on pre-frozen validation streams and distinguishes between-state disagreement from within-frozen-state actor disagreement. Failure of that probe removes the state-generation-variability interpretation even if the complete compiler method improves utility.

5 Evaluation Contract

The remaining program is staged so that an unfavorable result cannot be rescued by changing the estimand, choosing another state realization, opening an untouched benchmark, or adding a model after outcomes. Completed evidence, mechanism localization, method evaluation, and confirmatory transport remain separate authorities. The current M3R4 and Bridge V4-R2 designs have both passed independent pre-execution methodology review, but neither review grants provider execution authority.

5.1 M2: deterministic state-semantic sufficiency

Before testing an automatic generator intervention, M2 asks whether the manually identified repair surfaces can causally alter downstream behavior at all. Four frozen persistent states share the same base spreadsheet skill:

- G0: the exact initial skill;
- G1: G0 plus output save/reload/target-cell verification;
- G2: G0 plus a complete Inspect→Read→Compute→Write→Save→Reload/Verify loop;
- G3: G2 plus clean recovery from malformed or failed tool calls.

No updater is called. Each state is evaluated on the same 18 development held-out tasks with DeepSeek-V4-Pro, temperature zero, thinking disabled, $K = 1$, ten turns, and the same deterministic verifier. The primary screen selects the simplest G1<G2<G3 whose utility exceeds G0 by at least $1/18$. Historical S1 outcomes are excluded from the gate. If the primary passes, a pre-frozen 17-task sensitivity excluding the one task split across provider quota eras must remain strictly positive; this sensitivity may veto but never rescue.

M2 establishes at most *semantic sufficiency*. G1–G3 were manually constructed after the selected historical state was inspected, so they do not show that an automatic learner can discover or reliably synthesize the useful repair. The current Recovery V3 object remains frozen and exactly-once; its partial scientific outcome is not used in the manuscript.

5.2 M3R4: FULLY PROSPECTIVE SAME-EVIDENCE STATE-REALIZATION LOCALIZATION

The exact-evidence replay created two different free-form persistent states from reconstructed byte-identical learner-visible First-Fail evidence. Earlier M3 proposals reused outcome-consumed actor observations or mixed noncommensurate disagreement quantities. M3R4 supersedes them before any M3 scientific execution and keeps the total new actor budget fixed.

The localization uses exactly two already-existing persistent artifacts:

- FF_R1, SHA `596bd30b49...fdf04f`;
- FF_R2, SHA `fb5454a27f...36615e`.

Historical FF_HIST and WIN_COMMON remain descriptive background only. They receive no new M3R4 actor calls and do not enter the M3R4 gate.

On the same fixed outcome-selected 18-task development panel, M3R4 generates exactly two *new post-freeze* actor realizations for each state and task: 2 states $\times 18$ tasks $\times 2$ replicates $= 72$ actor units, with zero updater calls. Historical exact-replay actor outcomes do not count as any of these four observations. For task q, denote the new outcomes (A_1, A_2) for FF_R1 and (B_1, B_2) for FF_R2.

Define

$$D_X^{\mathrm{M3}}(q) = \tfrac{1}{4}\left(|A_1 - B_1| + |A_1 - B_2| + |A_2 - B_1| + |A_2 - B_2|\right), \tag{6}$$

$$D_A^{\mathrm{M3}}(q) = \tfrac{1}{2}\left(|A_1 - A_2| + |B_1 - B_2|\right), \tag{7}$$

$$E_{\mathrm{REAL}}^{\mathrm{M3}} = \mathrm{mean}_q D_X^{\mathrm{M3}}(q) - \mathrm{mean}_q D_A^{\mathrm{M3}}(q). \tag{8}$$

If the two persistent artifacts are byte-identical, $E_{\mathrm{REAL}}^{\mathrm{M3}}$ is defined as zero. Under conditionally iid, independent, stationary Bernoulli actor realizations within each frozen state/task block,

$$\mathbb{E}[E_{\mathrm{REAL}}^{\mathrm{M3}}] = \mathrm{mean}_q [p_A(q) - p_B(q)]^2. \tag{9}$$

Without that stochastic model, $E_{\mathrm{REAL}}^{\mathrm{M3}}$ remains an observed cross-state-minus-within-state disagreement statistic only.

Exact conditional uncertainty gate. For each task, condition on the total number of successes among (A_1, A_2, B_1, B_2). Under the task-wise equal-propensity null and within-task iid model, a task with exactly two successes has six equiprobable state-label allocations: two put both successes in one state and four split them. Let n_2 be the number of such informative tasks and X the number with both successes in one state. The aggregate law

$$X \mid n_2, \{\text{task totals}\} \sim \mathrm{Binomial}(n_2, 1/3) \tag{10}$$

is used only if the informative task-block indicators additionally satisfy a frozen *cross-task conditional factorization* qualification. This assumption does not require equal task-specific success probabilities or exchangeability of task identities; arbitrary shared provider/runtime shocks or other cross-task coupling block the exact Binomial interpretation.

A bounded selected-case localization requires all of: distinct state SHAs, $E_{\mathrm{REAL}}^{\mathrm{M3}} > 0$, one-sided $p_{\mathrm{exact}} \leq 0.05$, qualified within-task iid/stationarity, and qualified cross-task conditional factorization. If an inference assumption fails, the observed disagreement may be reported descriptively but cannot receive the inferential localization label, and no automatic rerun is authorized. M3R4 changes no task, state, model, or actor budget relative to M3R3. Its frozen design passed independent GPT-5.6 Sol / Extra High pre-execution review; execution still requires a separate contract and authorization.

5.3 M4: FRESH EVIDENCE$\times$GENERATOR BRIDGE

The automatic bridge uses a newly generated development suite disjoint from the selected S1 panel and from the untouched E3 confirmation lane. It contains 120 formal task artifacts: 96 update tasks arranged as 12 eight-task streams, 12 SCREEN held-out tasks, and 12 different VALIDATION held-out tasks. SCREEN and VALIDATION each contain one stream per controlled failure family, and their held-out panels are disjoint.

For each stream we freeze the balanced 2×2:

Evidence package	Native free-form updater	Typed compiler
Winner	W_FREE	W_COMP
First-Fail-4	FF4_FREE_A	FF4_COMP

Evidence semantics. Winner exposes the served winner trajectory+score package on all eight update pools. First-Fail-4 changes exactly four pre-hashed mixed pools to the deterministic first failed nonwinner while the other four retain the Winner package. FF4 therefore fixes a four-of-eight replacement count, not trajectory length, failure severity, semantic density, or information amount. The rejected-source contrast is an evidence-package-policy effect rather than a purified "failure versus success" treatment.

Information parity and free-form realization. Within each evidence package, FREE and COMP receive the same selected trajectory bytes and binary score. The compiler cannot read hidden arm, family, task, rollout, golden-answer, or held-out labels. Corresponding learner-input hashes must match before provider execution. The updater and actor use temperature zero, thinking disabled, and provider retry zero, but a reliable generation seed is not exposed for every call. The first free-form state is therefore a prospectively deployed realization. No additional draw may replace an unfavorable primary state. On VALIDATION only, exactly one additional byte-identical-evidence state, FF4_FREE_B, is pre-authorized regardless of A's outcome; B is sensitivity/mechanism only and never replaces A.

Universal state identity. Persistent state is the treatment. Before actor evaluation, *all* FREE, COMP, and generic-control artifacts are partitioned by full skill SHA-256. Any byte-identical arms share the same actor observation schedule, and every state-treatment contrast between identical SHAs is exactly zero. This applies to FREE↔COMP, COMP↔generic, FREE↔generic, and FREE_A↔FREE_B collisions. Calls removed by aliasing cannot be reallocated elsewhere.

5.4 FOUR ORTHOGONAL BRIDGE QUESTIONS

Bridge V4-R2 adjudicates four questions independently rather than one conjunctive PASS bit.

Q1: complete state-generation method effect (primary). For stream s, define

$$G_W(s) = J_s(\text{W\_COMP}) - J_s(\text{W\_FREE}), \tag{11}$$

$$G_{F,A}(s) = J_s(\text{FF4\_COMP}) - J_s(\text{FF4\_FREE\_A}), \tag{12}$$

$$G_{\text{MAIN},A}(s) = \tfrac{1}{2}\left[G_W(s) + G_{F,A}(s)\right]. \tag{13}$$

The equal weighting is frozen because the primary question is the generator factor in a balanced 2×2, not one evidence source. SCREEN passes Q1 only if mean $G_{\text{MAIN},A} > 0$, at least 4/6 SCREEN streams have $G_{\text{MAIN},A} > 0$, and there is no technical, evidence-parity, SHA-accounting, exactly-once, or protocol violation. Only this Q1 gate may open VALIDATION. VALIDATION uses the identical first-deployed-realization estimand and directional rule on six different streams and 12 disjoint tasks; FF4_FREE_B never enters the primary gate.

Q2: FF4-specific method substance. The two generic controls are explicitly evidence-source-specific: they compare with FF4_COMP under the FF4 evidence package and cannot explain or mediate the Winner-side contribution to balanced Q1. `SCORE_ONLY_GENERIC_MAX` receives the FF4 ordered score vector and no trajectory text. `SCOPE_MATCHED_GENERIC_MAX` additionally receives only k, the number of nonredundant repair blocks that FF4_COMP would render, but no trajectory text or primitive identity. It is trajectory-text-blind but diagnosis-cardinality-informed. Beating both controls supports FF4 trajectory-conditioned typed diagnosis beyond these frozen generic explanations; failure narrows only Q2 to scope/generic canonicalization and cannot erase an independently passed balanced Q1 effect.

Q3: FF4 same-evidence state-realization localization. On six pre-frozen VALIDATION tasks, ordinary validation supplies actor replicate 1 and exactly one additional replicate is pre-authorized for FF4_FREE_A, FF4_FREE_B, and FF4_COMP. No third execution exists. For FREE outcomes (A_1, A_2, B_1, B_2) define

$$D_X(s) = \tfrac{1}{4}\,\mathrm{mean}_q\left[|A_1 - B_1| + |A_1 - B_2| + |A_2 - B_1| + |A_2 - B_2|\right], \tag{14}$$

$$D_A(s) = \tfrac{1}{2}\,\mathrm{mean}_q\left[|A_1 - A_2| + |B_1 - B_2|\right], \tag{15}$$

$$E_{\mathrm{REAL}}(s) = D_X(s) - D_A(s). \tag{16}$$

A directional development localization requires mean $E_{\mathrm{REAL}} > 0$ and positive E_{REAL} on at least 4/6 VALIDATION streams. Unconditionally, this supports only observed cross-state disagreement exceeding within-state actor disagreement. The stronger identity $\mathbb{E}[E_{\mathrm{REAL}}] = \mathrm{mean}_q[p_A(q) - p_B(q)]^2$ is invoked only under the explicitly stated conditionally iid/independent stationary actor model given state/task/model/runtime. Logs can support but cannot prove that model; if it is not qualified, the observed directional Q3 result remains descriptive and the squared-propensity interpretation is dropped. If A and B are byte-identical, state-SHA aliasing forces $E_{\mathrm{REAL}} = 0$.

Q4: rejected-source moderator. Independently report

$$S_C(s) = J_s(\text{FF4\_COMP}) - J_s(\text{W\_COMP}), \tag{17}$$

$$I(s) = G_{F,A}(s) - G_W(s). \tag{18}$$

These quantities test First-Fail-4 source superiority and Evidence×Generator moderation. They are parked secondary hypotheses and cannot veto Q1–Q3.

5.5 Prespecified FREE-realization sensitivity

On VALIDATION only, report

$$G_{F,AB}(s) = J_s(\text{FF4\_COMP}) - \tfrac{1}{2}[J_s(\text{FF4\_FREE\_A}) + J_s(\text{FF4\_FREE\_B})], \tag{19}$$

$$G_{\mathrm{MAIN},AB}(s) = \tfrac{1}{2}[G_W(s) + G_{F,AB}(s)]. \tag{20}$$

These two-realization quantities are sensitivity analyses only. They never replace $G_{\mathrm{MAIN},A}$, never gate Q1, and cannot rescue a failed primary result. If the first-deployment effect passes but the sensitivity weakens materially or changes sign, the paper reports realization sensitivity rather than switching estimands.

5.6 Decision table and STOP rule

- If Q1 fails on SCREEN, VALIDATION remains sealed. If Q1 fails on VALIDATION, stop the automatic state-generation-method story. Do not rescue either failure with FF4_FREE_B, generic controls, Q3, Q4, E3, another backbone, or a public benchmark.
- If Q1 passes but the FF4 generic controls match the compiler, retain the complete generator-method effect while narrowing Q2 to scope/generic canonicalization. Do not claim FF4 trajectory-conditioned typed diagnosis.
- If Q1 and the relevant FF4 method-substance controls pass but Q3 fails or its stochastic model is not qualified, retain the complete typed-compiler result while dropping the stronger state-realization/propensity interpretation.
- If Q1–Q3 are supported but Q4 fails, retain the state-generation paper and explicitly drop rejected-evidence superiority. This is a clean primary-paper outcome rather than a failed Q1.
- Only a frozen fresh result surviving its claim-specific controls may motivate a separately reviewed untouched E3 proposal. Nothing in M4 itself grants E3 authority.

Bridge V4-R2 has passed independent GPT-5.6 Sol / Extra High pre-execution methodology review. This qualifies the frozen design only; it does not authorize search-pool acquisition, updater execution, actor evaluation, SCREEN opening, VALIDATION opening, or E3.

5.7 Scientific unit

For M3R4, the fixed selected panel is conditioned upon and task-level observations are not promoted to population samples. For M4, the independent scientific unit is the updated stream, not the held-out

Evidence source alone is unresolved. 48-pair MRW−WIN-C is small and heterogeneous.
⇓
Simple witness selection fails. Progress-based selectors do not recover the strong First-Fail case.
⇓
Freeze the state. The historical strong state remains directionally useful across two fresh actor remeasurements.
⇓
Freeze the evidence, regenerate the state. Two byte-identical-evidence updater realizations fail to recover the historical advantage.
⇓
Open causal question. Is this persistent-state regeneration instability beyond downstream actor noise, and can a controlled generator method improve fresh-state utility?

Figure 1: Current evidence ladder. Each step removes a simpler explanation but the completed evidence remains a selected-case regeneration anomaly rather than a population variance decomposition.

task executions or provider calls nested within a stream. The controlled suite is a mechanism substrate rather than external-validity evidence. No outcome-conditioned additional stream, state synthesis, actor replicate, backbone, or benchmark is permitted to rescue a failed frozen gate.

6 Current Results and Open Claim Gates

At this manuscript checkpoint, the large-sample evidence-source study, S1 selector pilot, historical frozen-state stability study, and exact-evidence replay are complete and outcome-readable. M2 Recovery V3 is an active frozen measurement object whose incomplete outcomes remain sealed. The revised M3 regeneration audit and M4 automatic bridge are prospective. We report only completed evidence here.

6.1 Rejected-witness evidence is not a stable universal treatment

The 48-pair DeepSeek V2 study does not pass its frozen superiority criterion. The mean rejected-witness-minus-winner contrast is $+0.0231$, with bootstrap interval $[-0.0185, 0.0660]$ and one-sided sign-flip $p = 0.1719$. The stream distribution includes positive, null, and negative effects. This directly narrows the paper: the central contribution cannot be "failed trajectories are better learning data."

6.2 A diagnostic-witness selector fails its development gate

On the selected `e1-tsr-00` stream, First-Fail is 17/18 while WIN-C is 13/18, but progress-based failure and progress-contrast states are both 14/18. Because the prospective question was whether a more progress-diagnostic selector repairs the instability, the result is a failure of that hypothesis rather than evidence for First-Fail as a general rule. No S2 selector expansion is opened.

6.3 State freezing and evidence replay point to a regeneration anomaly

The historical strong First-Fail state remains directionally stronger than its WIN-C comparator when both states are frozen: 15/18 versus 14/18 in one complete remeasurement and 16/18 versus 12/18 in a second. The same historical advantage does not survive state regeneration from byte-identical First-Fail evidence. Fresh updater realization 1 yields 15/18 versus a fresh WIN-C run of 15/18; realization 2 yields 11/18 versus 12/18.

Figure 1 summarizes what these observations do and do not remove. Byte-identical evidence rules out evidence disappearance. Repeated evaluation of the historical strong state makes a pure one-lucky-actor-roll explanation less plausible. But regression-to-the-mean, selected-case bias, evaluator alignment, and fresh-comparator/actor variability remain. The completed result is therefore *state-regeneration instability in one controlled development case*, not a measured state-generation variance component.

6.4 THE STRONGEST CURRENT CLAIM

The strongest manuscript-level statement supported today is:

> In one controlled development case, reconstructed byte-identical trajectory evidence did not reliably regenerate the historical behaviorally useful skill through the native free-form updater, while the historical state itself remained directionally useful when frozen and re-evaluated. This observation is consistent with a persistent-state generation bottleneck but does not by itself isolate generator variance from downstream actor noise or selected-case effects.

The manuscript does not currently claim that the typed compiler improves downstream utility, that First-Fail-4 is superior to winner evidence, that state-generation variability dominates actor variability, or that the mechanism generalizes beyond the controlled development substrate.

6.5 WHAT THE REVISED NEXT GATES DECIDE

M2 remains a semantic-sufficiency probe for the manually identified G1–G3 surfaces. The current M3R4 design instead concentrates the unchanged 72-unit actor budget on two fully post-freeze replicates of each of the already-existing FF_R1 and FF_R2 states across the fixed 18-task panel, with zero updater calls. Historical actor outcomes and WIN-C remain descriptive only. Its primary observed localization uses cross-state disagreement minus within-frozen-state actor disagreement; the stronger squared-propensity and exact conditional interpretations are allowed only when both within-task iid/stationarity and cross-task conditional-factorization qualifications hold. M3R4 has passed independent pre-execution review but has not been executed.

M4 no longer defines the generator question through the First-Fail cell alone. Bridge V4-R2 makes the equal-weight generator-factor main effect across the predeclared Winner and First-Fail-4 evidence cells primary, using the first prospectively deployed FREE state in the FF4 cell. First-Fail source superiority and the Evidence$\times$Generator interaction remain independent moderators. The generic controls are explicitly FF4-specific: they determine whether an FF4 compiler advantage supports trajectory-conditioned typed diagnosis or only scope/generic canonicalization, and they cannot explain or veto the Winner-side contribution to balanced Q1. A same-evidence FF4 repeated-synthesis probe on VALIDATION reports observed cross-state minus within-state actor disagreement; a success-propensity interpretation is conditional on the stated iid/stationarity model. V4-R2 has independently passed pre-execution design review but has not been executed.

If the fresh generator-factor effect fails its frozen SCREEN/VALIDATION path, the automatic state-generation-method story stops without E3, another backbone, or another benchmark as rescue. If Q1 passes but an FF4 generic control matches FF4_COMP, the raw balanced complete-method effect is retained while the FF4 method-substance interpretation narrows to scope or generic canonicalization; this cannot explain away the Winner-side component of Q1. If Q1 passes but First-Fail source/interaction fails, the paper becomes the cleaner state-generation-method paper and explicitly drops failure-evidence superiority. If the Bridge realization probe fails, or its iid/stationarity model is not qualified, the paper drops the stronger state-realization/propensity mechanism language without rewriting the separately adjudicated Q1 result.

7 DISCUSSION, LIMITATIONS, AND CONCLUSION

The scientific shift is from evidence selection to state regeneration. The initial E2-R17 hypothesis emphasized which rejected trajectories should reach persistent learning. The closed large-sample study and selector intervention do not support a universal witness rule. The more informative observation is downstream: one persistent state is useful when frozen, yet the same updater does not reliably recreate that historical advantage from identical evidence. This changes the primary scientific question from “which failure should we show?” to “what happens when the same evidence is materialized into persistent state again?”

The completed result is instability, not a variance decomposition. The exact-evidence replay contains only two fresh updater realizations in one outcome-selected development stream, and each fresh state was originally observed through a fresh downstream actor execution. Frozen-state remeasurement of the historical strong state partially addresses actor noise, but the fresh FF1/FF2 states have not yet been evaluated through a fully prospective repeated-actor design. We therefore avoid claiming a measured generator variance component or that updater variance dominates actor variance. M3R4 targets this residual uncertainty without another updater call by allocating the same 72 actor units to two new post-freeze replicates of each fresh state and by separating observed excess disagreement from model-based propensity inference. Its exact conditional p-value additionally requires cross-task factorization; failure of that qualification blocks the inferential label rather than triggering a rerun.

The compiler is an intervention before it is an algorithmic contribution. Trace-conditioned diagnosis, success/failure contrast, execution-grounded repair, and persistent knowledge accumulation already appear in SkillRevise, SkillCAT, RethinkSkill, and WikiSkill (Liu et al., 2026a; Chen et al., 2026; Liu et al., 2026b; Tang et al., 2026). Three spreadsheet-oriented primitives plus deterministic rendering are not, by themselves, a convincing universal algorithm. Their role in the present paper is to create a fully specified alternative generator under matched evidence. A broader algorithmic claim would require fresh evidence that the controlled generator itself provides a nontrivial advantage beyond generic workflow instructions and state scope.

Generic workflow hygiene and state scope are load-bearing alternatives. Completion, verification, and recovery are sensible spreadsheet-agent instructions independent of any rejected trajectory. Moreover, a trajectory-conditioned compiler may simply emit fewer clauses than a maximal generic checklist. The revised bridge therefore includes a score-only generic maximum and a trajectory-text-blind, diagnosis-cardinality-informed scope control. These controls are deliberately *FF4-specific*: they test what explains an FF4 compiler advantage and cannot explain the Winner-side contribution to the balanced generator main effect. If FF4_COMP cannot beat the scope-matched control, the FF4 trajectory-conditioned diagnosis claim fails; a separately passed Q1 complete-method effect can remain, but its method-substance interpretation is correspondingly narrower.

Failure evidence is no longer a gate on the primary paper. The large-sample evidence already shows that rejected-witness effects are heterogeneous. Accordingly, the revised bridge makes the balanced generator-factor main effect across Winner and First-Fail-4 evidence the primary estimand, while treating source superiority and the Evidence×Generator interaction as secondary moderators. The primary method effect can therefore validate even if First-Fail is no better than Winner or if the compiler benefit is not concentrated in the FF4 cell. In that case, the cleaner paper is about the persistent-state generation interface rather than failure-evidence selection.

The controlled suite is not external validity. The six failure families are generated to permit matched causal interventions and deterministic verification. They should not be interpreted as a representative sample of user spreadsheet workflows. SpreadsheetBench and later workflow-level benchmarks can test transport only after the internal mechanism survives; opening them after a negative fresh generator result would be benchmark shopping.

Selected-case development remains a source of researcher degrees of freedom. The S1 stream and the semantics used to design G1–G3 were selected after historical outcomes. We treat them as a mechanistic sandbox rather than independent confirmation. M3R4 is fully prospective with respect to its 72 actor observations, but it remains conditional on that historically selected stream and 18-task panel, so it cannot erase selected-case bias. The fresh bridge breaks this adaptive loop more directly by freezing new tasks, stream assignment, held-out panels, compiler vocabulary, generic controls, state-identity rules, and decision gates before bridge outcomes.

Practical implication. Persistent text state is attractive because it is inspectable, but inspectability does not imply reproducibility. Self-evolution studies should therefore report the identity of realized persistent artifacts, not only the trajectory source and final task score. Content-addressed state IDs, same-evidence regeneration checks, and frozen-state actor remeasurement make it possible to distinguish an evidence-policy question from a state-materialization question.

Conclusion. Our current evidence rejects a simple universal story that failed trajectories are better training signals and reveals a narrower phenomenon: in one controlled development case, a historically useful persistent skill remains behaviorally active when frozen, while reconstructed byte-identical evidence does not reliably regenerate the same advantage through the native updater. This is state-regeneration instability consistent with a state-generation bottleneck, not yet a population variance decomposition. We therefore make the state generator an explicit experimental variable, retarget the next frozen-state audit to existing same-evidence states, and prospectively compare native free-form generation with a typed canonical intervention in a balanced Evidence$\times$Generator design. The primary test is the generator-factor effect itself; generic controls, state-realization localization, and rejected-source moderation independently determine how that effect may be interpreted. A failed primary generator result is not rescued by a new failure selector, a larger benchmark, or another backbone.

Reproducibility Statement

All load-bearing controlled experiments bind task IDs, initial and learned skill SHA-256 values, trajectory evidence hashes, model identity, provider budgets, retry rules, and completion leases. Completed units are not replayed to repair scientific outcomes. M3R4 binds two fully prospective actor replicates per frozen state/task and separates within-task iid/stationarity from cross-task factorization as explicit inference qualifications. The bridge suite is generated deterministically from content-addressed salts, with disjoint SCREEN and VALIDATION panels; V4-R2 collapses byte-identical persistent states into universal SHA equivalence classes, preserves one primary generator-factor estimand from SCREEN through VALIDATION, and keeps FF4 method-substance, state-realization localization, and rejected-source moderation orthogonal to that gate.

AI Use Statement

Generative AI tools were used to assist with hypothesis refinement, research software, literature search, adversarial methodology review, manuscript drafting, and artifact inspection. Quantitative claims are checked against persisted experiment receipts and content-addressed artifacts. The authors are responsible for the final scientific claims, citations, and submission.

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Table 2: Current manuscript claim ledger after independent manuscript review.

Status	Statement
Supported	The 48-pair rejected-witness effect is small and heterogeneous under the frozen DeepSeek V2 design.
Supported	On the selected development stream, the historical strong First-Fail persistent state remains directionally useful across two frozen-state actor remeasurements.
Supported	Two fresh free-form updater states generated from reconstructed byte-identical First-Fail evidence do not reproduce the historical First-Fail advantage in their original measurements.
Supported, local	The completed observations show a state-regeneration anomaly consistent with a persistent-state generation bottleneck and rule out evidence disappearance as a sufficient explanation.
Not supported	Rejected failures are universally superior learning evidence.
Not supported	Progress proximity or diagnosticity defines a validated witness-selection law.
Not supported	Generator variance has been isolated from actor variance or estimated at population scale.
Not supported yet	Typed compilation improves downstream utility or beats trajectory-blind state-scope-matched generic construction.
Not supported yet	The regeneration phenomenon generalizes to fresh streams, untouched E3, a second backbone, or a public benchmark.

A Claim Boundary

B Frozen Evidence Summary

DeepSeek V2 Repair2. The frozen design contains 48 pairs, 96 learned states, and 1,728 held-out units. Mean MRW−WIN-C is 0.023148, median 0.013889, standard deviation 0.077970, one-sided sign-flip $p = 0.171875$, and bootstrap interval $[-0.018519, 0.065972]$. The adjudicated verdict is `HOLD_MRW_UNDERPOWERED_OR_HETEROGENEOUS`.

Single-case witness selector. On `e1-tsr-00`, WIN-C/First-Fail/Progress-Fail/Progress-Contrast are 13/18, 17/18, 14/18, and 14/18. The frozen verdict is `S1_SIGNAL_FAIL_STOP_NO_S2`.

Historical frozen-state stability. Two complete remeasurements give First-Fail versus WIN-C of 15/18 versus 14/18 and 16/18 versus 12/18. Verdict: `FIRST_FAIL_FROZEN_STATE_STABILITY_PASS`.

Exact-evidence updater replay. All reconstructed learner-visible packet hashes match the historical First-Fail evidence. Fresh updater realization 1 gives 15/18 versus a fresh WIN-C run of 15/18; realization 2 gives 11/18 versus 12/18. The historical First-Fail state SHA is `97e28b4862...185252`, the two fresh First-Fail state SHAs are `596bd30b49...fdf04f` and `fb5454a27f...36615e`, and all three available WIN-C updater artifacts collapse to the same persistent-state SHA `6df40f6170...c00649`. These completed results remain historical context; M3R4 does not reuse their actor outcomes in its new gate.

C Prospective Stage Status

M2. The deterministic G0–G3 state-semantic intervention is frozen under the existing exactly-once Recovery V3 object. Its incomplete scientific outcome remains sealed at this manuscript checkpoint. M2 tests manual semantic sufficiency only.

M3R4 pre-execution object. M3R4 is the current superseding M3 design and has passed independent pre-execution review. It uses only the two already-existing fresh same-evidence states FF_R1 and FF_R2 and allocates the unchanged 72 new actor units as two fully post-freeze replicates per state$\times$task on the fixed 18-task panel; updater calls remain zero. Historical FF_HIST, WIN_COMMON, and original exact-replay actor outcomes are descriptive only. The observed localization is $E^{\mathrm{M3}}_{\mathrm{REAL}} = D^{\mathrm{M3}}_{X} - D^{\mathrm{M3}}_{A}$. A stronger exact conditional state-label result additionally requires: distinct state SHAs, positive observed excess, $p_{\mathrm{exact}} \leq 0.05$, qualified within-task iid/stationarity, and qualified conditional factorization across informative task blocks. The latter makes $X \mid n_2, \{\text{task totals}\} \sim \mathrm{Binomial}(n_2, 1/3)$ valid without assuming equal task probabilities or exchangeable task identities. Failure of either stochastic qualification blocks inference rather than authorizing a rerun.

Bridge V4-R2 pre-execution object. The fresh Evidence×Generator bridge has also passed independent pre-execution methodology review and remains zero-provider. The primary Q1 estimand is the equal-weight generator-factor main effect $G_{\mathrm{MAIN},A} = \frac{1}{2}[(W_{\mathrm{COMP}} - W_{\mathrm{FREE}}) + (FF4_{\mathrm{COMP}} - FF4_{\mathrm{FREE},A})]$, frozen identically at SCREEN and VALIDATION. First-Fail source superiority and the Evidence×Generator interaction are independent moderators and cannot veto Q1. The score-only and diagnosis-cardinality-informed generic controls are explicitly ***FF4-specific***: they classify FF4 typed-diagnosis substance but cannot explain the Winner-side contribution to balanced Q1. On VALIDATION only, one additional same-evidence FF4 FREE state per stream is reserved for sensitivity/localization and cannot replace the first state. State-SHA equivalence is universal across FREE, COMP, and generic arms, forcing zero state-treatment/realization contrasts for byte-identical artifacts. Bridge Q3 supports observed excess disagreement unconditionally; the squared-propensity interpretation additionally requires the frozen conditionally iid/independent stationary actor model.

M5. Untouched E3 confirmation, a second backbone, and public benchmarks have no authority. Failure of the corrected fresh Q1 generator-factor path stops the automatic state-generation-method story without those rescue expansions. Failure of a generic control or realization/moderator subtest instead narrows the corresponding interpretation and cannot be used to switch the Q1 estimand.

D Bridge Suite Construction

The fresh bridge suite uses new controlled blocks 7–9 and has zero task-ID overlap with the earlier 378-task controlled suite. Blocks 7 and 8 provide update candidates; block 9 provides held-out candidates. From 162 deterministic candidates, hash-based selection freezes 96 update tasks in 12 eight-task streams, 12 SCREEN held-out tasks, 12 disjoint VALIDATION held-out tasks, 12 update integrity reserves, and 30 held-out integrity reserves. Each SCREEN/VALIDATION stream set contains one stream per failure family; each held-out panel contains two tasks per family. Reserved tasks may replace only a pre-execution file-integrity failure, never a model failure or unfavorable scientific outcome.